NammaSpace: Code Repository

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1. Repository

https://github.com/chepuridathathreya/Namma-Space (public)

2. Contents

Table I
Repository layout

Path	Contents
`README.md`	Overview, block diagram, setup and run instructions
`run_all.py`	One script for the whole pipeline on one machine (`--check`, `--test`, `--video`)
`namma_pipeline/`	Prepare, train and finish scripts and the viewer template
`test/sample_scene.zip`	Sample data for the test run: 40 frames with the camera model and collision map already solved
`docs/`	Capture SOP, run guide and how the system works
`01_room`, `02_hallway_bedroom`, `03_bedroom_early`	The three published viewers

3. Setup and Rebuild

```
git clone https://github.com/chepuridathathreya/Namma-Space
cd Namma-Space
pip install -r requirements.txt
python run_all.py --check          # lists anything missing
python run_all.py --test           # bundled sample, short training
python run_all.py --video my_walkthrough.mp4
```

Requirements: Linux or WSL2, an NVIDIA GPU with 16 GB or more, 20 GB of free disk, ffmpeg, COLMAP 3.9 or newer, and nerfstudio with gsplat (`requirements-train.txt`). The README on the repository front page explains every command.

4. Status

Tested: preparation and finishing on Windows; training on Linux in our two-machine setup; the requirement check and the finishing step on the bundled sample.

Not yet tested: the one-machine training run (`--test`) on a separate clean GPU machine.